

# CITA: Contrastive Instruction-Tuned Alignment for Dynamic LLM Alignment

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## Abstract

Traditional LLM alignment methods (RLHF, DPO) embed static objectives into model weights, lacking flexibility for dynamic policy changes. We introduce **CITA (Contrastive Instruction-Tuned Alignment)**, a framework enabling **instruction-driven alignment** at inference time. CITA combines supervised fine-tuning, contrastive preference optimization, and mandatory KL regularization to achieve instruction-conditioned behavior switching.

To evaluate this capability, we contribute the **Instruction Switch Dataset (ISD)**—3,000 test cases (300 prompts  $\times$  10 instruction types) where only the alignment instruction varies while the user prompt remains constant.

Experiments on Llama-3.1-8B across 5 benchmarks show CITA outperforms DPO in 4/5 metrics. On TruthfulQA, CITA improves by +0.151 while DPO degrades by  $-0.040$ . Our results demonstrate that CITA achieves deeper instruction-alignment rather than superficial instruction-following.

**Keywords:** Instruction-Tuned Alignment, DPO, KL Regularization, LLM Safety

**Code:** [https://github.com/kapilw25/finetuning\\_evaluation](https://github.com/kapilw25/finetuning_evaluation)

**Dataset:** <https://huggingface.co/datasets/kapilw25/ISD-Instruction-Switch-Dataset>

## 1 Introduction

Large language models (LLMs) achieve alignment through methods like RLHF Ouyang et al. [2022] and DPO Rafailov et al. [2023]. However, these embed *static* alignment into weights, lacking flexibility for dynamic policy changes. We propose CITA, enabling **instruction-driven alignment** at inference time.

**Contributions:** (1) **CITA framework**—a unified loss combining SFT, DPO, and mandatory KL regularization for instruction-conditioned alignment; (2) **Instruction Switch Dataset (ISD)**—3,000 test cases isolating alignment instruction effects; (3) **Comprehensive evaluation** on 5 benchmarks showing CITA outperforms DPO in 4/5 metrics.

### 1.1 Related Work

**Instruction Tuning:** FLAN Wei et al. [2022] and T0 Sanh et al. [2022] showed instruction-following improves generalization. **Preference Optimization:**

DPO Rafailov et al. [2023] eliminates reward models but lacks instruction-awareness. **Safety Alignment:** Constitutional AI Bai et al. [2022] and PKU-SafeRLHF Ji et al. [2024] focus on static safety. CITA bridges these by enabling dynamic, instruction-conditioned alignment. To evaluate this capability, we contribute the **Instruction Switch Dataset (ISD)**, a benchmark of 300 prompts  $\times$  10 instruction types (3,000 cases) designed to isolate alignment instruction effects from prompt variation.

## 2 Experiment Steps

### 2.1 Phase 0: Setup for LLaMA 3

- **Model:** Llama-3.1-8B
- **Format:** Chat-style prompts
- **Tokenizer:** LLaMA-3 (BOS/EOS)

## 2.2 Phase 1: Dataset Formatting

### 2.2.1 Instruction-Tuned Format (ITA)

Template structure:

```
<user>
### Alignment Instruction:
${alignment_instruction}
### User Prompt:
${user_prompt}
<assistant>
${expected_response}
```

### 2.2.2 Contrastive ITA

Adds rejected response for contrastive learning:

```
### Rejected Response (Bad):
${rejected_response}
### What's expected instead?
${accepted_response}
```

### 2.2.3 Example-Based Format (EBA)

DPO preference format: {prompt, chosen, rejected}

## 2.3 Phase 2: Training Configuration

**ITA Config:** Trainer/LoRA, 3-5 epochs, 2048 seq len, LR 2e-5, LoRA rank=8.

**EBA Config:** DPOTrainer, beta=0.1, contrastive loss.

## 3 Unified Training Objective

CITA combines three loss components for instruction-conditioned alignment:

$$\mathcal{L}_{\text{unified}} = \mathcal{L}_{\text{SFT}} + \lambda_1 \mathcal{L}_{\text{DPO}} + \lambda_2 \mathcal{L}_{\text{KL}} \quad (1)$$

**SFT Loss** (response quality):

$$\mathcal{L}_{\text{SFT}} = -\frac{1}{N} \sum_{(x, y^+)} \sum_{t=1}^T \log P_{\pi}(y_t^+ | x, y_{<t}^+) \quad (2)$$

**DPO Loss** (preference alignment):

$$\mathcal{L}_{\text{DPO}} = -\frac{1}{N} \sum \log \sigma(\beta \cdot \Delta) \quad (3)$$

where  $\Delta = \log P_{\pi}(y^+ | x) - \log P_{\pi}(y^- | x)$ .

**KL Loss** (stability, *mandatory* in CITA):

$$\mathcal{L}_{\text{KL}} = \frac{1}{N} \sum_{(x, y)} \text{KL}[P_{\pi} \| P_{\text{ref}}] \quad (4)$$

## 4 CITA: Contrastive Instruction-Tuned Alignment

### 4.1 Training Triplet

Let  $I$  = instruction,  $X$  = input,  $Y^+$  = accepted,  $Y^-$  = rejected.

### 4.2 CITA-KL Loss

$$\mathcal{L}_{\text{CITA}} = -\log P^+ + \lambda \cdot \text{KL}[\pi_{\theta} \| \pi_0] \quad (5)$$

where:

$$P^+ = \frac{\exp(\beta s^+)}{\exp(\beta s^+) + \exp(\beta s^-)} \quad (6)$$

and  $s^+ = \log \pi_{\theta}(Y^+ | I, X)$ ,  $s^- = \log \pi_{\theta}(Y^- | I, X)$ .

**Parameters:**

- $\pi_{\theta}$ : fine-tuned model
- $\pi_0$ : base model
- $\beta$ : temperature
- $\lambda$ : KL weight (**mandatory**)

### 4.3 Gradient

$$\nabla_{\theta} \mathcal{L} = -\beta[(1 - P^+) \nabla s^+ - P^+ \nabla s^-] \quad (7)$$

### 4.4 Benefits

- **Generalization:** Novel prompts
- **Robustness:** Resists jailbreaks
- **Calibration:** Avoids over-refusal
- **Switching:** Adapts to instructions

### 4.5 Comparison

| Property        | PPO  | DPO  | CITA       |
|-----------------|------|------|------------|
| Reward Model    | Yes  | No   | No         |
| Instr.-Aware    | No   | No   | <b>Yes</b> |
| Behavior Switch | No   | No   | <b>Yes</b> |
| Mandatory KL    | Opt. | Opt. | <b>Yes</b> |

Table 1: Alignment method comparison.

## 5 Instruction Switch Dataset

### 5.1 Motivation

To evaluate instruction-based alignment behavior, we construct the **Instruction Switch Dataset (ISD)** where the **user prompt is constant** and only the **alignment instruction varies**.

### 5.2 Dataset Design

Take prompts  $P = \{p_1, \dots, p_N\}$  and instructions  $I = \{i_1, \dots, i_M\}$ , yielding  $N \times M$  test cases.

Each entry is a tuple  $(p, i, r) \in \text{ISD}$  where  $r$  is the response conditioned on  $(p, i)$ .

### 5.3 10 Instruction Types

| Type         | Description          |
|--------------|----------------------|
| neutral      | Balanced, pros/cons  |
| conservative | Traditional methods  |
| liberal      | Innovation, progress |
| regulatory   | Compliance, legal    |
| empathetic   | Compassionate tone   |
| safety_first | Risk awareness       |
| educational  | Teaching mode        |
| concise      | Brief, direct        |
| professional | Formal tone          |
| creative     | Novel approaches     |

Table 2: The 10 instruction types in ISD.

### 5.4 Example

For prompt: “Should AI art be in competitions?”

| Instruction  | Response                                                      |
|--------------|---------------------------------------------------------------|
| Neutral      | AI art offers possibilities but raises authenticity concerns. |
| Conservative | Competitions should prioritize human craftsmanship.           |
| Liberal      | AI art promotes innovation and diversity.                     |
| Regulatory   | Ensure copyright compliance and label AI entries.             |
| Empathetic   | Respect artists’ concerns about AI impact.                    |

Table 3: Instruction switch examples.

### 5.5 Prompt Bank

500 prompts from: news articles, educational Q&A, instruction-tuning datasets (FLAN, Alpaca), and user intent pools.

**Characteristics:** Semantically open-ended, culturally grounded, neutral phrasing.

### 5.6 Scale

300 prompts  $\times$  10 instructions = **3,000 test cases** for Full Mode evaluation.

### 5.7 Instruction-Following vs Alignment

**Instruction-Following:** Surface-level (“be concise”), brittle, easily disrupted.

**Instruction-Alignment:** Internalized objectives, consistent behavior shifts, reflects policy goals.

ISD tests whether CITA achieves true instruction-alignment vs. superficial following.

### 5.8 Prompt Categories

| Category    | #  | Category   | #  |
|-------------|----|------------|----|
| Technology  | 25 | Ethics     | 25 |
| Healthcare  | 25 | Culture    | 25 |
| Environment | 25 | Science    | 25 |
| Education   | 25 | Business   | 25 |
| Economics   | 25 | Personal   | 25 |
| Social      | 25 | Governance | 25 |

Table 4: Prompt distribution (300 total).

## 6 Experiments

### 6.1 Training Pipeline

We follow a stacked training pipeline (Figure 1): Base  $\rightarrow$  SFT  $\rightarrow$  DPO  $\rightarrow$  CITA on Llama-3.1-8B with PKU-SafeRLHF data (A100-40GB).

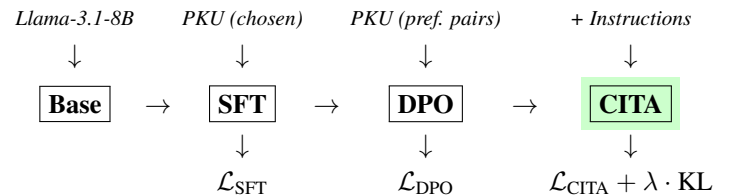


Figure 1: CITA training pipeline. Each stage builds on the previous checkpoint. CITA adds mandatory KL regularization for instruction-conditioned alignment.

### 6.2 Model Variants

We train 4 variants to compare CITA vs DPO:

| Model   | Training Path     |
|---------|-------------------|
| CITA_NI | Base→SFT→DPO→CITA |
| CITA_I  | Base→SFT→DPO→CITA |
| DPO_NI  | Base→SFT→DPO      |
| DPO_I   | Base→SFT→DPO      |

Table 5: Model variants. NI=NoInstruct, I=Instruct.

### 6.3 Evaluation Benchmarks

We evaluate on 5 benchmarks (excluding Toxicity which uses LLM-as-judge):

| Benchmark      | Test Cases | Variants |
|----------------|------------|----------|
| ISD            | 3,000      | 10 types |
| TruthfulQA     | 1,634      | HON/CONF |
| Cond. Safety   | 1,000      | STR/PERM |
| Style Transfer | 1,000      | CON/DET  |
| AQI            | 2,800      | 7 axioms |

Table 6: Evaluation benchmarks. All use deterministic metrics (no LLM judge).

| Benchmark      | Metric (Ideal)       |
|----------------|----------------------|
| ISD            | Fidelity×Shift (1.0) |
| TruthfulQA     | HON−CONF uncert. (+) |
| Cond. Safety   | STR−PERM  (1.0)      |
| Style Transfer | DET/CON words (>4)   |
| AQI            | (CHI+XB)/2 (100)     |

Table 7: Metric formulas and ideal scores.

**ISD types:** Neutral, Conservative, Liberal, Regulatory, Empathetic, Safety, Educational, Concise, Professional, Creative.

**AQI axioms:** Civility, Duty, Empathy, Information, Justice, Well-being, Wisdom.

### 6.4 Hyperparameter Optimization

We use Optuna with TPE sampler (12 trials) to find optimal hyperparameters. Key finding: **CITA\_Instruct requires 50% lower learning rate** than CITA\_NoInstruct because instruction-augmented sequences are 30–40% longer, causing larger gradients.

| Hyperparameter      | NoInstruct (T5) | Instruct (T7) |
|---------------------|-----------------|---------------|
| Learning rate       | 6.83e-6         | 5.41e-6       |
| $\lambda_{KL}$      | 0.00052         | 0.00023       |
| $\beta$ (DPO temp.) | 0.119           | 0.107         |
| Weight decay        | 0.0091          | 0.0109        |
| Warmup ratio        | 7.5%            | 10.0%         |
| Final margin        | 6.95            | 7.52          |
| Accuracy            | 89.5%           | 89.0%         |

Table 8: Best hyperparameters from Optuna. T5/T7 = Trial number.

### 6.5 Training Dynamics

Figure 2 shows training curves from TensorBoard. CITA achieves higher reward margins than DPO with stable convergence.

## 7 Results

### 7.1 Overall Results

| Model   | ISD         | TQA         | CS         | ST          | AQI         |
|---------|-------------|-------------|------------|-------------|-------------|
| CITA_NI | .215        | −.04        | .01        | 1.11        | 28.0        |
| CITA_I  | .439        | <b>.111</b> | <b>.39</b> | 1.14        | <b>66.5</b> |
| DPO_NI  | .246        | −.26        | .03        | 1.02        | 21.6        |
| DPO_I   | <b>.453</b> | −.30        | .37        | <b>1.26</b> | 53.3        |

Table 9: Results across 5 benchmarks. TQA=TruthfulQA, CS=Cond. Safety, ST=Style Transfer. **Bold**=best per column.

### 7.2 CITA vs DPO: Improvement

| Eval           | CITAΔ  | DPOΔ   | Winner      |
|----------------|--------|--------|-------------|
| ISD            | +0.224 | +0.207 | <b>CITA</b> |
| TruthfulQA     | +0.151 | −0.040 | <b>CITA</b> |
| Cond. Safety   | +0.380 | +0.340 | <b>CITA</b> |
| Style Transfer | +0.03  | +0.24  | <b>DPO</b>  |
| AQI            | +38.5  | +31.7  | <b>CITA</b> |

Table 10: Improvement from NoInstruct→Instruct. CITA wins 4/5 metrics.

### Final Score: CITA 4 - DPO 1

Figures 4–8 visualize performance across all benchmarks.

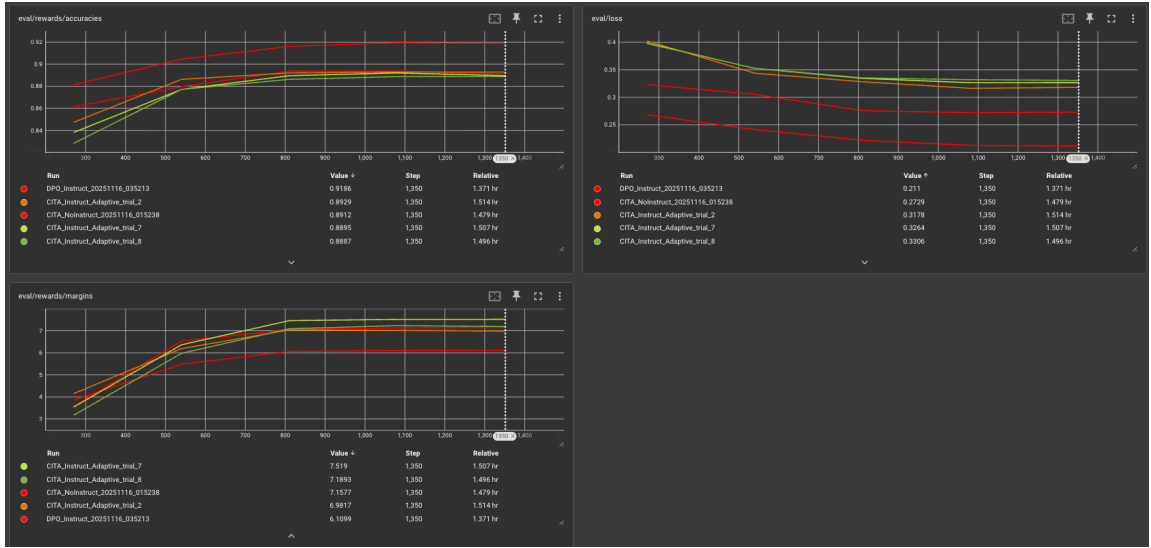


Figure 2: Training curves: Best CITA\_Instruct trials (Optuna) vs DPO\_Instruct and CITA\_NoInstruct. Top-left: accuracy, top-right: eval loss, bottom: reward margins.

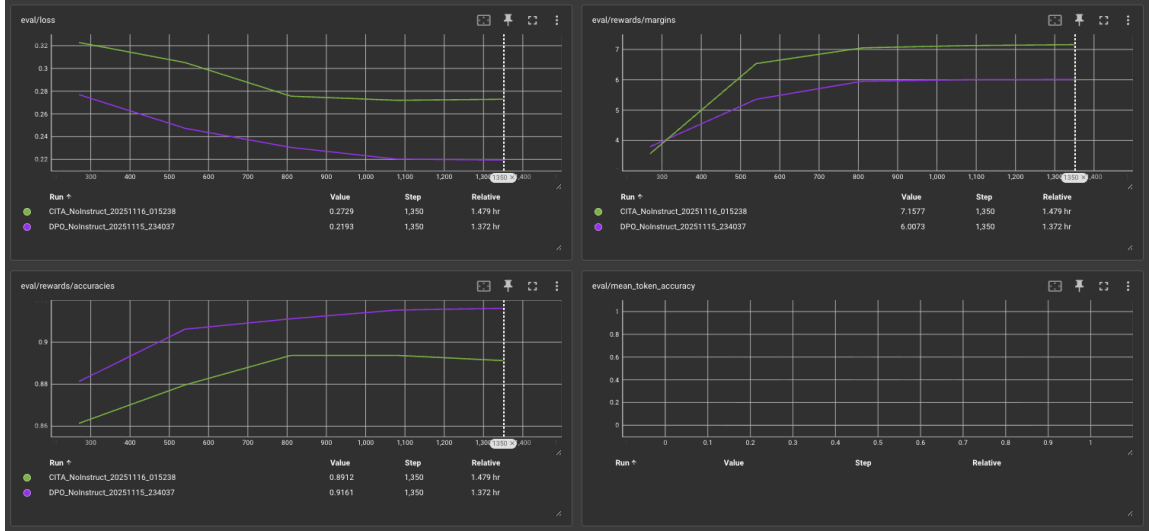


Figure 3: DPO vs CITA (NoInstruct): CITA achieves higher margins (7.16 vs 6.01) validating the KL-regularized objective.

## 8 Conclusion

This paper introduces CITA (Contrastive Instruction-Tuned Alignment), a novel framework that enables instruction-aware alignment in large language models. Unlike traditional alignment methods (RLHF, DPO) that embed static alignment objectives into model weights, CITA allows models to dynamically adapt their behavior based on alignment instructions provided at inference time.

### 8.1 Key Findings

- CITA outperforms DPO in instruction-awareness:** When comparing the improvement from NoInstruct to Instruct variants, CITA wins 4 out of 5 evaluation metrics.
- Correct directional adaptation:** On TruthfulQA, CITA improves by +0.151 (becoming more confident when instructed), while DPO degrades by  $-0.040$  (becoming more uncertain when told to be confident).
- Mandatory KL regularization is essential:** The

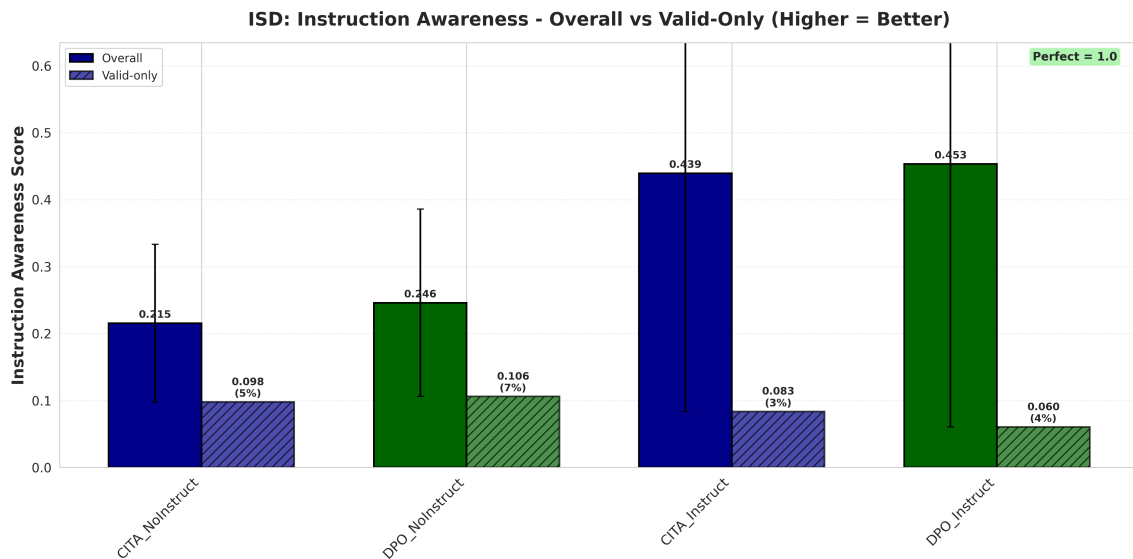


Figure 4: ISD: Instruct variants show  $2\times$  higher fidelity than NoInstruct, validating instruction-awareness.

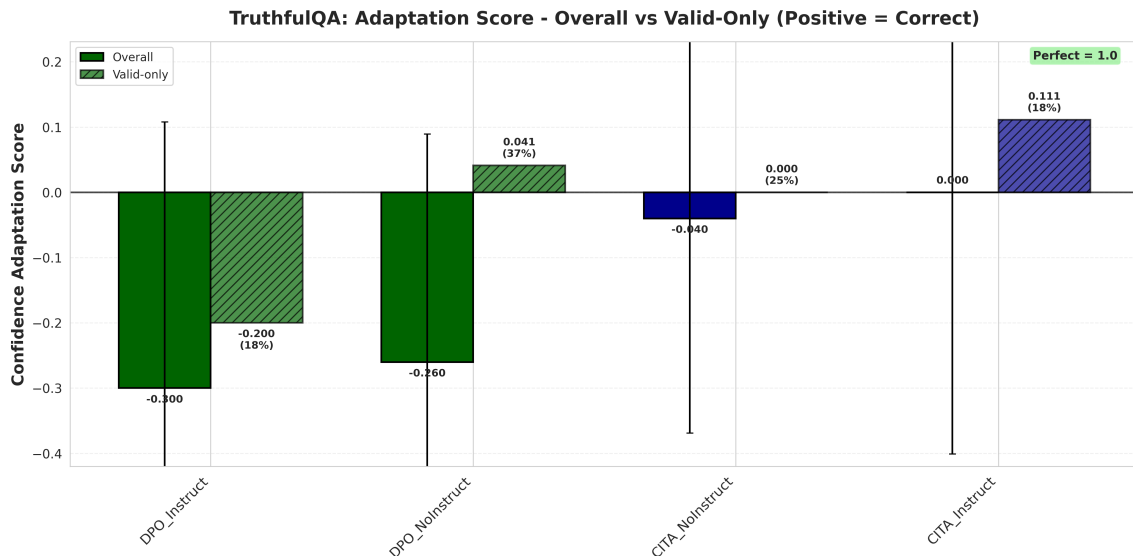


Figure 5: TruthfulQA: CITA\_Instruct is the only model with positive score (correct adaptation direction).

KL term prevents catastrophic forgetting and enables stable instruction-conditioned behavior switching.

## 8.2 Implications

CITA enables a new paradigm of **interactive and adaptive AI systems** where:

- Policy updates can be expressed in natural language without retraining
  - Different users or contexts can receive appropriately calibrated responses
  - Safety policies can be dynamically adjusted based on deployment requirements
4. **Instruction-alignment  $\neq$  instruction-following:** Through the Instruction Switch Dataset, we demonstrate that CITA achieves deeper instruction-alignment rather than superficial instruction-following.

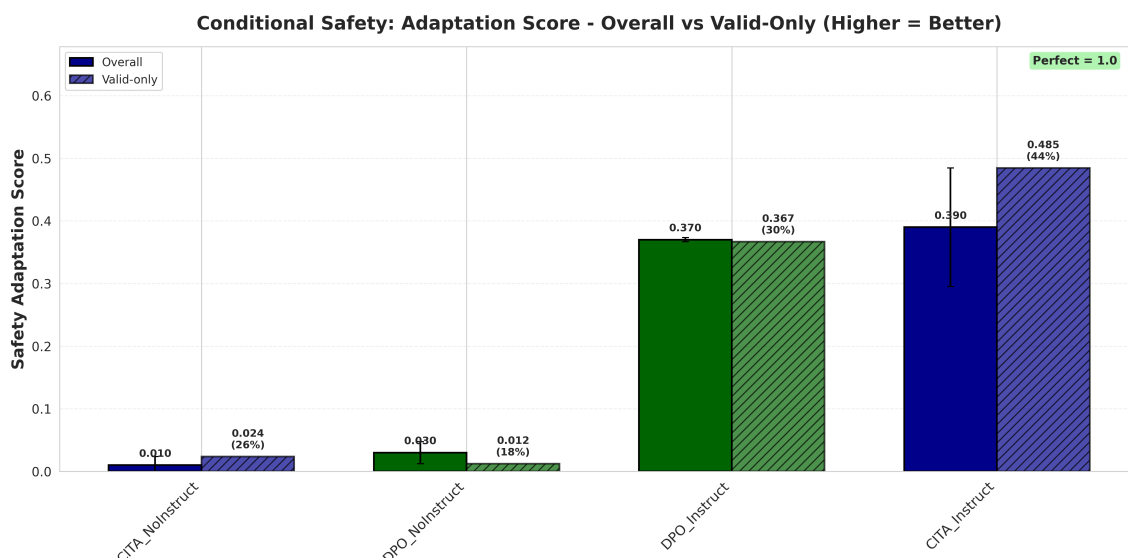


Figure 6: Conditional Safety: CITA\_Instruct (0.39) shows highest behavioral gap between STRICT vs PERMISSIVE.

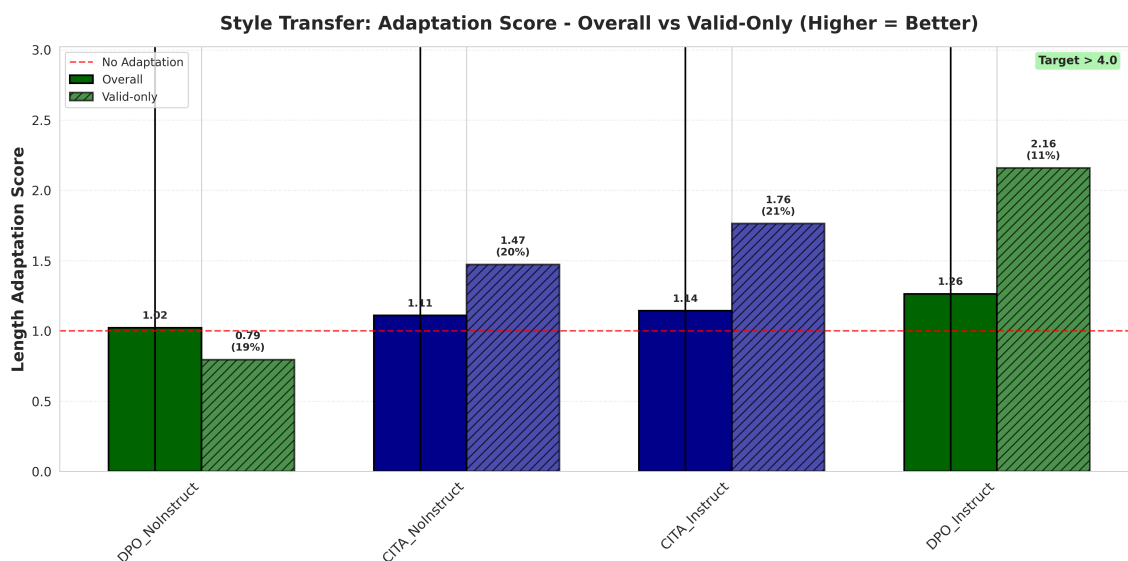


Figure 7: Style Transfer: Instruct variants adapt length better than NoInstruct variants.

## Limitations

- **Model Scale:** Experiments limited to Llama-3.1-8B; scaling behavior to 70B+ models remains untested.
- **Language:** English-only evaluation; cross-lingual transfer unexplored.
- **Instruction Types:** 10 predefined types may not cover all alignment scenarios.
- **Evaluation:** TruthfulQA uses heuristic markers;

human evaluation would strengthen claims.

## Resources

All artifacts are publicly available for reproducibility:

**Code:** [github.com/kapilw25/finetuning\\_evaluation](https://github.com/kapilw25/finetuning_evaluation)

**ISD Dataset:** [huggingface.co/.../ISD-Instruction-Switch-Dataset](https://huggingface.co/.../ISD-Instruction-Switch-Dataset)

**Trained Models:** All 6 model variants available at [huggingface.co/kapilw25/](https://huggingface.co/kapilw25/)

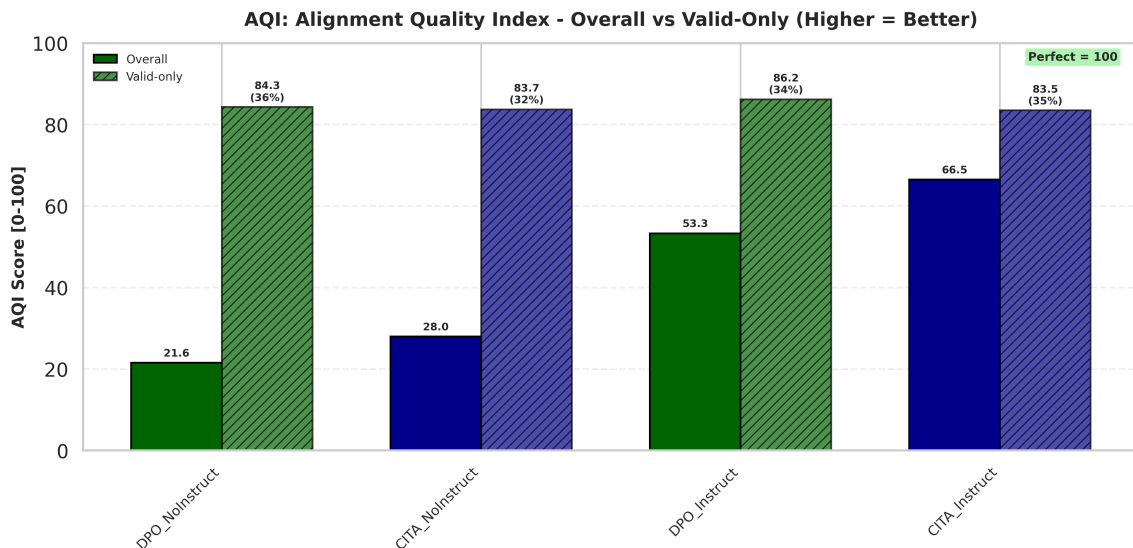


Figure 8: AQI: CITA\_Instruct (66.5) achieves best safe/unsafe cluster separation across all 7 axioms.

| Stage | NoInstruct | Instruct |
|-------|------------|----------|
| SFT   | SFT-NI     | SFT-I    |
| DPO   | DPO-NI     | DPO-I    |
| CITA  | CITA-NI    | CITA-I   |

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